TRAN QUANG DUYEN

AI Engineer · LLM Applications · RAG & AI Agents

tranquangduyen2503@gmail.com | +84 398 116 991 | github.com/duyentq-aie | Portfolio: duyentq-aie.github.io | Vinh Long, Vietnam

SUMMARY

IT graduate from FPT University who builds LLM features for e-commerce. My main project is a RAG + LangGraph shopping agent for an online store, where I owned the whole workflow: preparing and embedding product data, hybrid vector search, a controlled agent loop, LLM API integration, and measuring output quality before tuning prompts. I ship these as FastAPI services with Docker and CI, track latency and cost per conversation, and explain results in plain terms for non-technical teammates. 4-month backend internship on an AWS product, working in Agile sprints with an English-speaking client team.

SKILLS

AI / LLM: ML/DL & LLM fundamentals (embedding, tokenization, context window), RAG, AI agents, prompt engineering
Frameworks: LangChain, LangGraph, vector databases (Qdrant, ChromaDB)
Backend & APIs: Python, FastAPI, REST API, MongoDB, PostgreSQL
DevOps & cloud: Docker, CI/CD (GitHub Actions), AWS, Git
Computer vision: OpenCV, YOLO

PROJECTS

ShopMate — RAG + AI agent sales assistant for an e-commerce store

10/2026

Personal project · github.com/duyentq-aie/shopmate

Python · LangGraph · Gemini · Qdrant · bge-m3 · Saleor (GraphQL) · FastAPI · Langfuse · Docker · GitHub Actions

- Built a shopping agent that answers product, return-policy and order-status questions for a Saleor e-commerce store (250 products), using the store's GraphQL API as agent tools.
- Engineered the agent loop for production: read-only tools, max 8 steps per turn, support tickets only after the customer confirms, and a code-level guardrail that blocks any reply quoting a wrong price.
- Raised search accuracy for Vietnamese queries typed without accents by **+0.43 MRR** (95% CI 0.27–0.58) by fusing accent-folded keyword search with bge-m3 vectors in Qdrant.
- Measured output quality before tuning prompts: 104-question golden set, test split locked and run once: **31/33** correct answers, intent routing 33/33, **2.6 s** median latency, **~US$0.9 per 1,000 conversations**; every run traced in Langfuse, 1,600+ tests in CI.

DevPrep — AI mock-interview platform

2026

Capstone project (team) · github.com/Usopp204/DevPrep · FPT University

.NET 8 · FastAPI · Gemini · ChromaDB · InsightFace · SpeechBrain · Deepgram · Docker

- Built RAG-based answer grading: Gemini scores each answer against reference answers retrieved from ChromaDB (bge-m3) on a 50/30/20 rubric (knowledge / problem-solving / communication).
- Integrated third-party AI APIs into one interview flow (Deepgram speech-to-text, Gemini hints, Piper text-to-speech), plus multimodal anti-cheating with face (ArcFace) and voice embeddings.
- Owned 3 of 10+ .NET 8 microservices and co-owned the FastAPI AI service, all running in Docker Compose.

SentinelAI — Traffic violation detection from camera video

10/2026

Personal project · github.com/duyentq-aie/SentinelAI

Python · PyTorch · YOLO11 · OpenCV · FastAPI · Kaggle

- Built an end-to-end computer vision system (YOLO11 trained on Kaggle GPU, OpenCV, object tracking) that detects traffic violations and streams results to a FastAPI web app.
- Found train/test leakage in the original data, re-split by video and reported honest results (test mAP50 0.93); on a new camera the retrained model **halved false alarms**, with a human review page for each alert.

WORK EXPERIENCE

Backend Developer Intern — BIWOCO Software Company Limited

10/2024 – 02/2025

Software for the Australian market · 4 months

Python · FastAPI · MongoDB · AWS · Microservices · REST API

- Developed async REST APIs with FastAPI and Pydantic, with MongoDB queries (pagination, filtering, projection).
- Shipped features and bug fixes in a microservice backend on AWS; kept OpenAPI/Swagger docs up to date.
- Worked in Agile sprints, joining English-language stand-ups and sprint reviews with the Australian client team.

EDUCATION, AWARDS & LANGUAGES

- **FPT University** — Information Technology, 2022 – 2026
- Top 15 — FPTU Can Tho Olympic AI 2026, "AI for Smart Mekong Delta" (AI hackathon)
- Practical AI Training Program — FPT University Can Tho, 07–08/2026
- **English** — reads technical documentation fluently; follows English-language meetings